Exploratory Data Analysis (EDA) Report – Netflix Dataset

1. Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on the Netflix dataset to uncover meaningful insights and answer business-related questions. The analysis helps in understanding content trends, user preferences, and platform growth over time.

2. Dataset Description

The dataset contains information about Netflix movies and TV shows, including:

- Title
- Type (Movie / TV Show)
- Director
- Country
- Date Added
- Release Year
- Rating
- Duration

The dataset was pre-cleaned and used for analysis.

3. Data Cleaning & Preprocessing

The following steps were performed to prepare the dataset:

- Handled missing values by replacing them with "Unknown" where necessary
- Converted date columns into proper datetime format
- Removed inconsistencies in categorical data
- Ensured correct data types for analysis

4. Univariate Analysis

Univariate analysis was performed to understand individual variables:

- **Content Type Distribution**
  Movies dominate the dataset compared to TV Shows
- **Ratings Analysis**
  Most content falls under a few common rating categories

- **Country-wise Distribution**
  Some countries contribute significantly more content than others

5. Multivariate Analysis

Relationships between multiple variables were explored:

- **Content Addition Trend Over Time**
  A steady increase in content addition is observed, especially after 2015
- **Content Type vs Duration**
  Movies and TV shows differ significantly in duration patterns
- **Correlation Analysis**
  Visualizations were used to identify relationships between features

6. SQL Business Insights

SQL queries were used to answer key business questions:

1. Top 5 countries with the highest number of titles
2. Monthly trend of content addition
3. Top 5 directors with the most content
4. Distribution of Movies vs TV Shows
5. Most common content ratings
6. Average duration by content type
7. Year with the highest number of titles added

7. Dashboard

A static dashboard was created using Excel to visualize key performance indicators (KPIs), including:

- Total content count
- Distribution of content types
- Country-wise contribution
- Monthly trends

This dashboard provides a quick and clear overview of important insights.

8. Key Insights

- Movies are more common than TV shows on Netflix

- The United States has the highest number of titles
- Content addition increased significantly after 2015
- Certain ratings and directors dominate the platform
- Trends show strategic growth in content over time

9. Conclusion

This project demonstrates how EDA and SQL can be used to extract meaningful insights from real-world datasets. The findings can help businesses understand content trends, optimize strategies, and make data-driven decisions.

10. Future Scope

- Perform predictive analysis on content trends
- Build interactive dashboards using Power BI or Tableau
- Analyze user behavior and recommendation systems